

# DataPilot AI - Data Science Report

Automated Machine Learning Analysis

## 1. Dataset Overview

|                     |                        |
|---------------------|------------------------|
| File Name           | employee_attrition.csv |
| Rows                | 20                     |
| Columns             | 11                     |
| Numeric Columns     | 4                      |
| Categorical Columns | 7                      |

## 2. Dataset Understanding

**Summary:** This dataset contains information about employees, including their ID, age, salary, years at the company, and various categorical attributes such as gender, department, education, work-life balance, overtime status, marital status, and attrition. The data has no missing or duplicate rows.

**Problem Type:** Classification

**Observations:**

- The dataset is small with only 20 rows, which might limit the model's ability to generalize well.
- There are four numeric features (EmployeeID, Age, Salary, YearsAtCompany) and seven categorical features (Gender, Department, Education, WorkLifeBalance, OverTime, MaritalStatus, Attrition).
- The target variable for classification is 'Attrition', which indicates whether an employee has left the company.

## 3. Data Quality

| Metric               | Value |
|----------------------|-------|
| Duplicate Rows       | 0     |
| Duplicate Percentage | 0.00% |
| Constant Columns     | 0     |
| Empty Columns        | 0     |

## 4. Data Cleaning

| Metric        | Value |
|---------------|-------|
| Original Rows | 20    |
| Final Rows    | 20    |

|                               |   |
|-------------------------------|---|
| Duplicate Rows Removed        | 0 |
| Columns Removed               | 0 |
| Missing Value Columns Handled | 0 |

#### Cleaning Summary:

- Trimmed whitespace from string columns.

## 5. Analysis Plan

| Target Column     | Attrition      |
|-------------------|----------------|
| Problem Type      | Classification |
| Evaluation Metric | f1_score       |
| Train/Test Split  | 0.2            |
| Random State      | 42             |
| Scaling           | standard       |
| Feature Encoding  | one_hot        |
| Target Encoding   | label          |

**Numerical Features:** Age, Salary, YearsAtCompany

**Categorical Features:** Gender, Department, Education, WorkLifeBalance, OverTime, MaritalStatus

## 6. Exploratory Data Analysis

### *Numerical Summary*

#### Age

| Statistic | Value  |
|-----------|--------|
| count     | 20.000 |
| mean      | 33.500 |
| std       | 6.460  |
| min       | 24.000 |
| 25%       | 28.750 |
| 50%       | 32.500 |
| 75%       | 38.250 |
| max       | 45.000 |

#### Salary

| Statistic | Value |
|-----------|-------|
|-----------|-------|

|       |            |
|-------|------------|
| count | 20.000     |
| mean  | 68500.000  |
| std   | 18343.219  |
| min   | 43000.000  |
| 25%   | 55750.000  |
| 50%   | 63000.000  |
| 75%   | 82500.000  |
| max   | 102000.000 |

**YearsAtCompany**

| Statistic | Value  |
|-----------|--------|
| count     | 20.000 |
| mean      | 6.750  |
| std       | 4.459  |
| min       | 1.000  |
| 25%       | 3.000  |
| 50%       | 5.500  |
| 75%       | 10.000 |
| max       | 16.000 |

***Categorical Summary***

**Gender**

| Category | Count |
|----------|-------|
| Male     | 10    |
| Female   | 10    |

**Department**

| Category | Count |
|----------|-------|
| IT       | 7     |
| Sales    | 5     |
| HR       | 4     |
| Finance  | 4     |

**Education**

| Category | Count |
|----------|-------|
| Bachelor | 9     |
| Master   | 8     |

|     |   |
|-----|---|
| PhD | 3 |
|-----|---|

#### WorkLifeBalance

| Category  | Count |
|-----------|-------|
| Good      | 9     |
| Excellent | 7     |
| Fair      | 4     |

#### OverTime

| Category | Count |
|----------|-------|
| No       | 12    |
| Yes      | 8     |

#### MaritalStatus

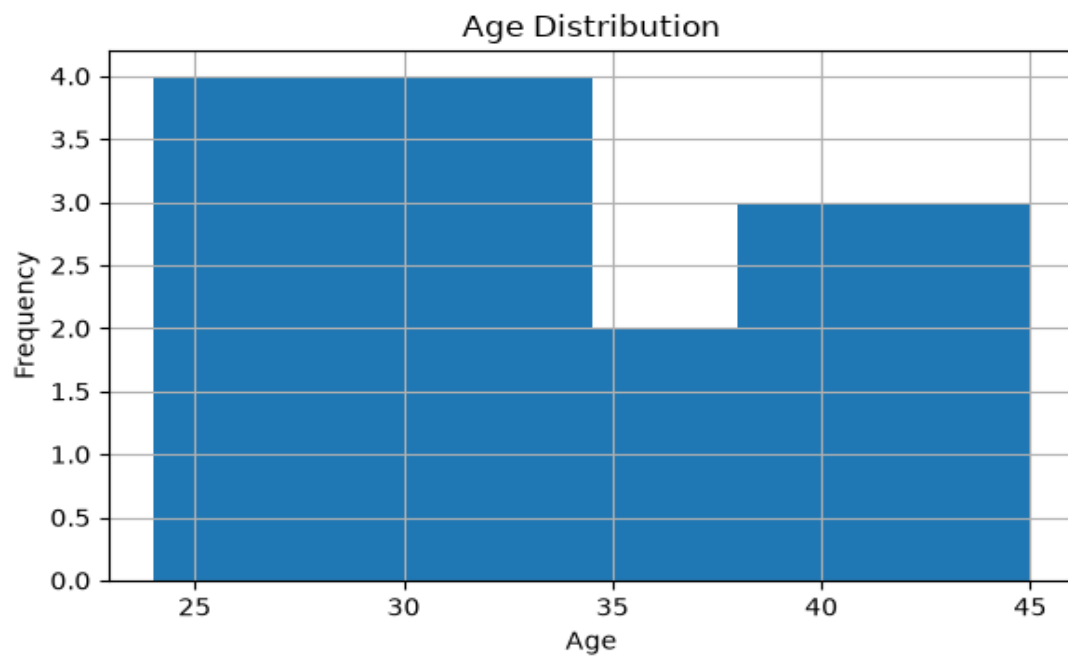
| Category | Count |
|----------|-------|
| Married  | 10    |
| Single   | 8     |
| Divorced | 2     |

#### *Target Distribution*

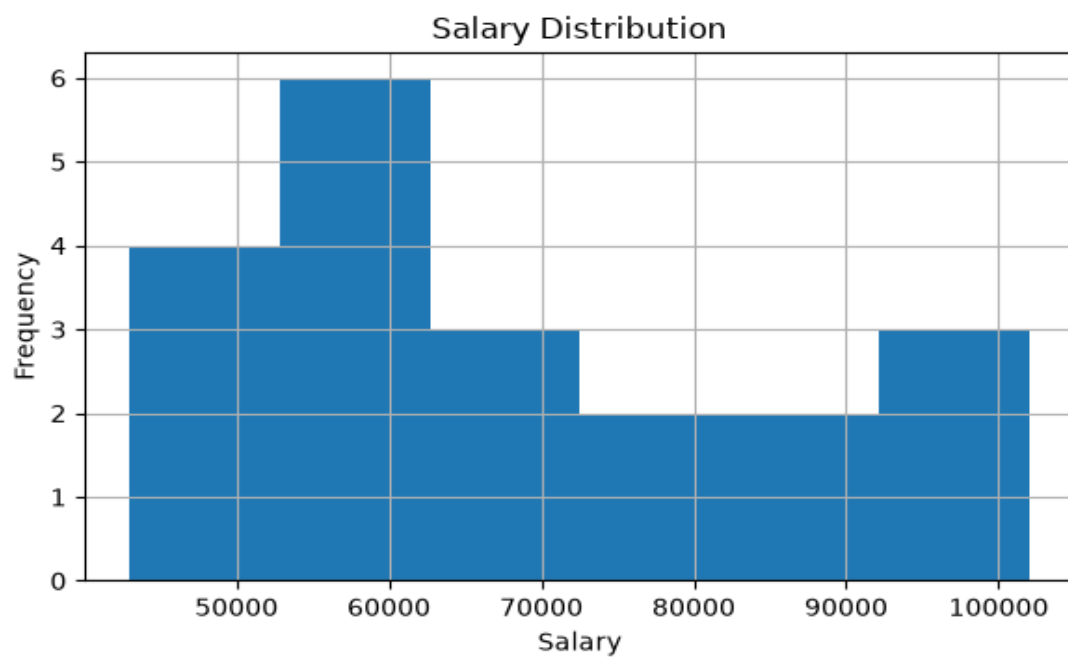
| Target | Count |
|--------|-------|
| No     | 12    |
| Yes    | 8     |

## 7. EDA Visualizations

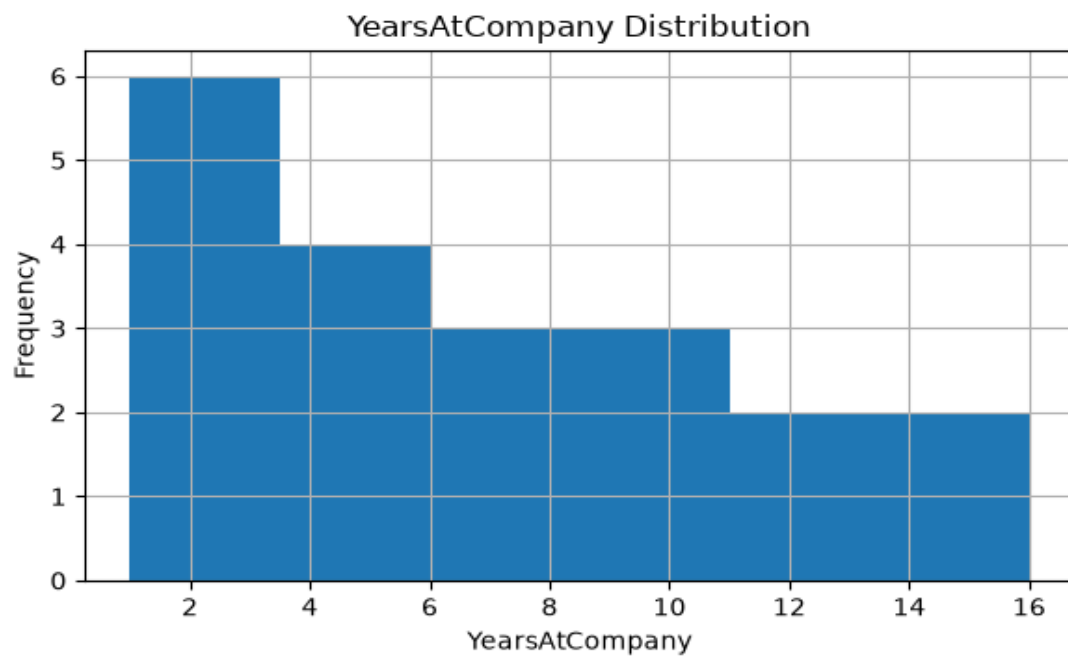
### *Age Histogram*



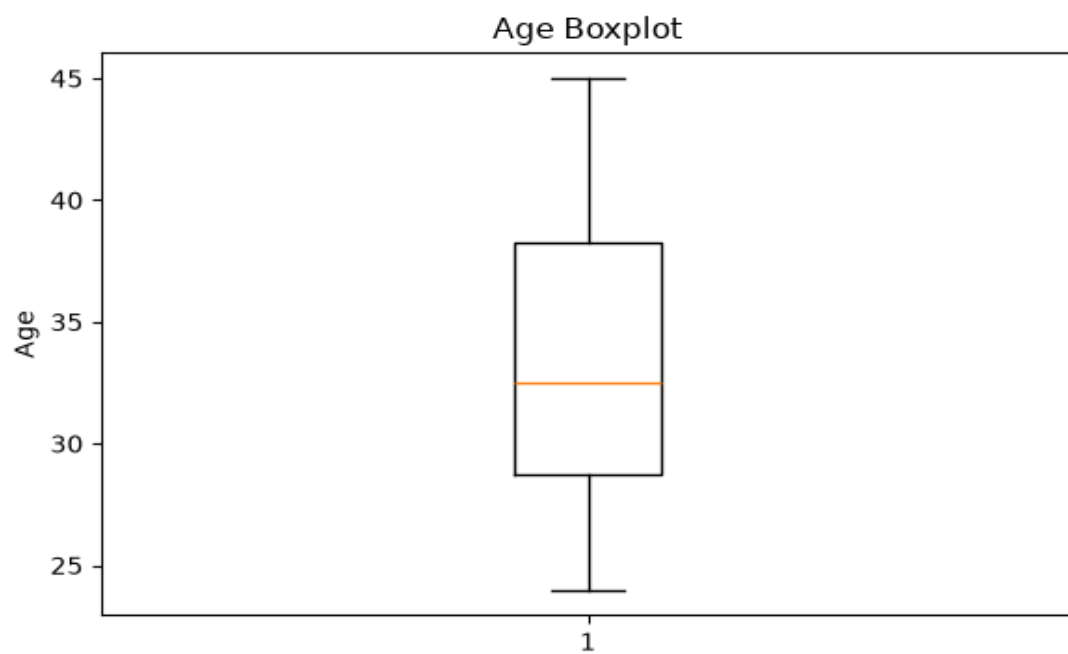
### *Salary Histogram*



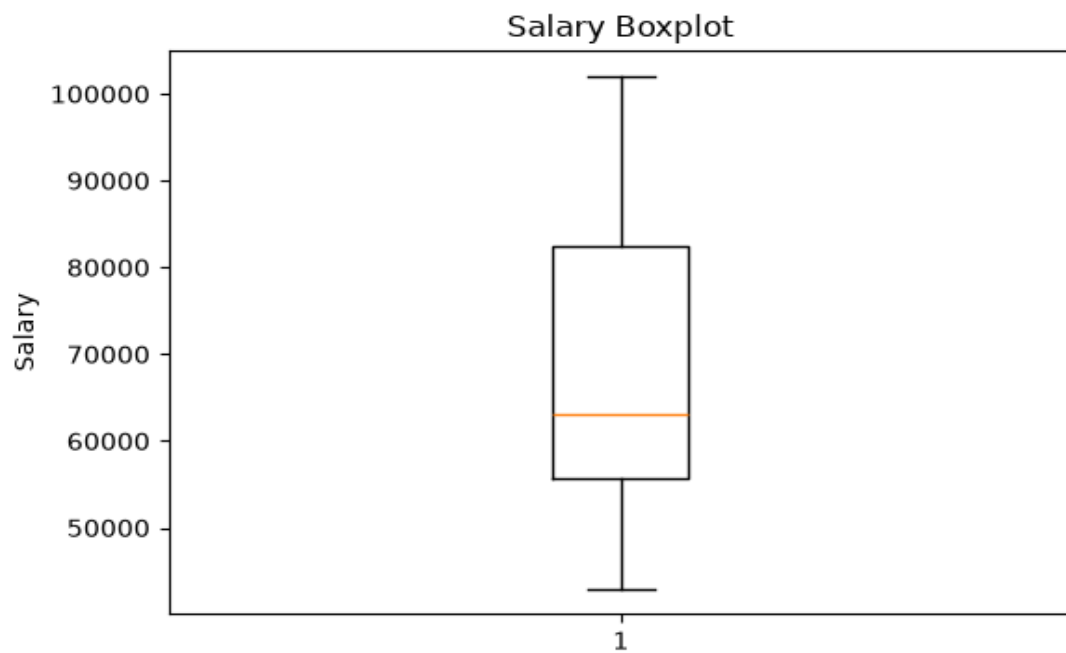
### *Yearsatcompany Histogram*



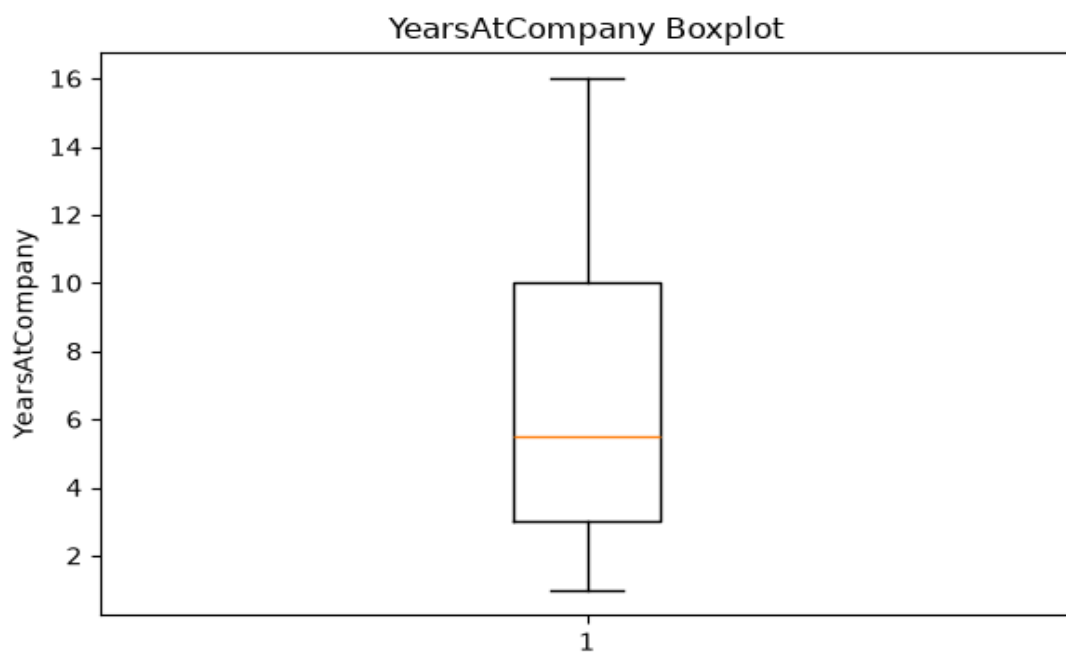
**Age Boxplot**



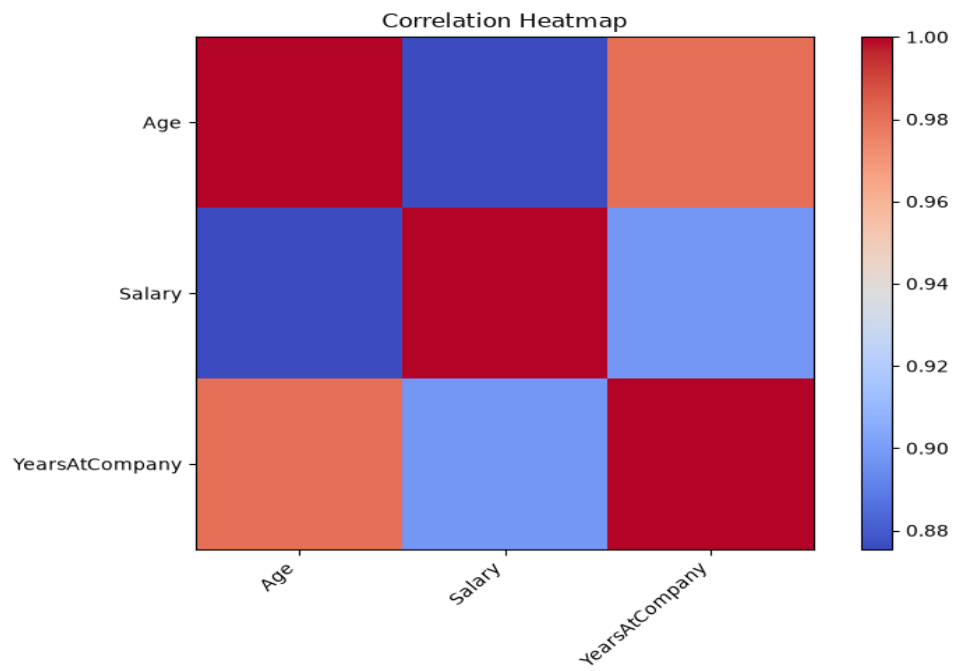
**Salary Boxplot**



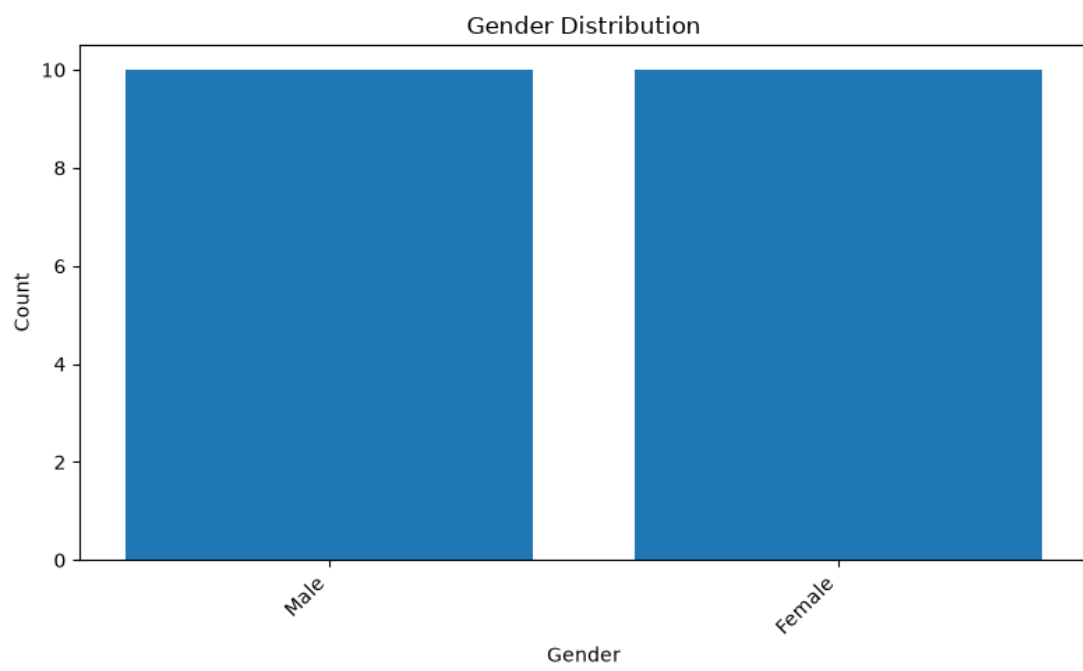
***Yearsatcompany Boxplot***



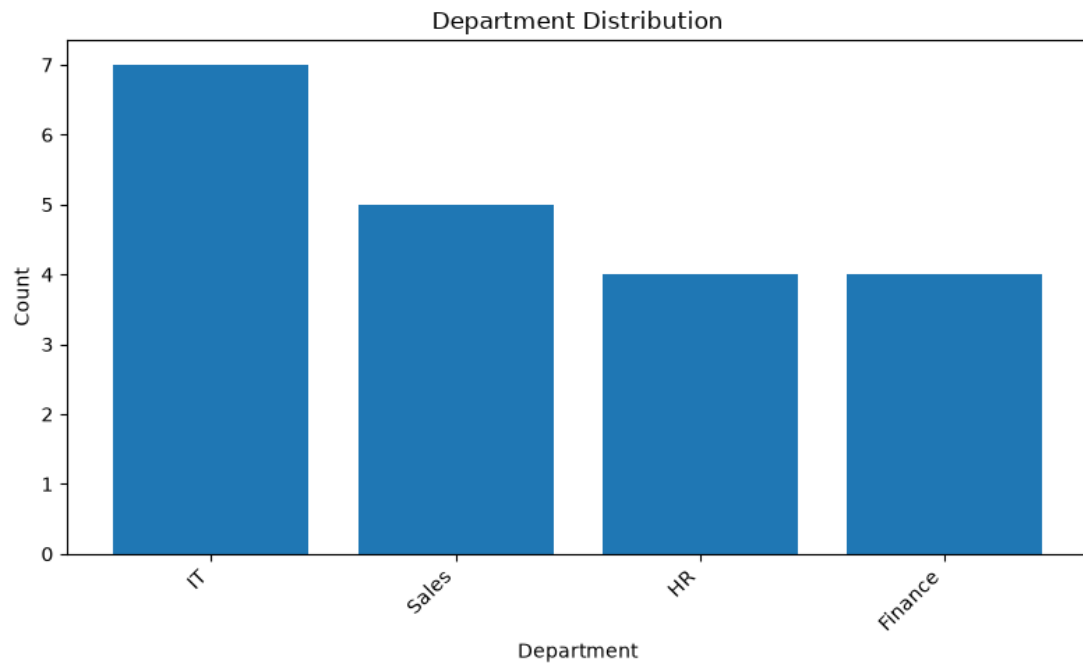
***Correlation Heatmap***



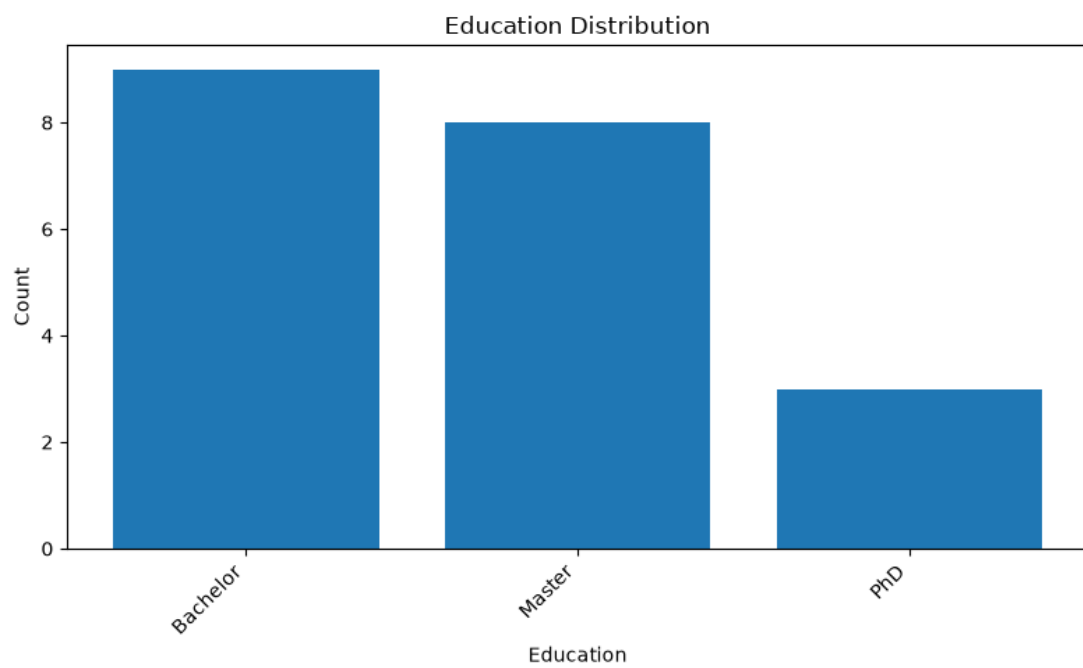
**Gender Countplot**



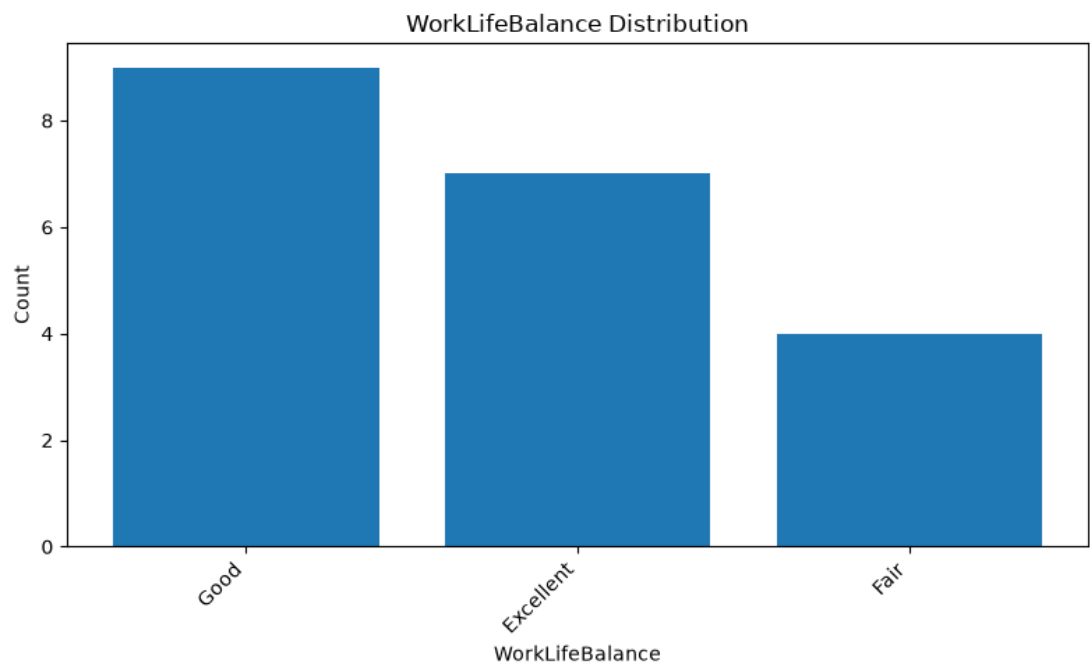
**Department Countplot**



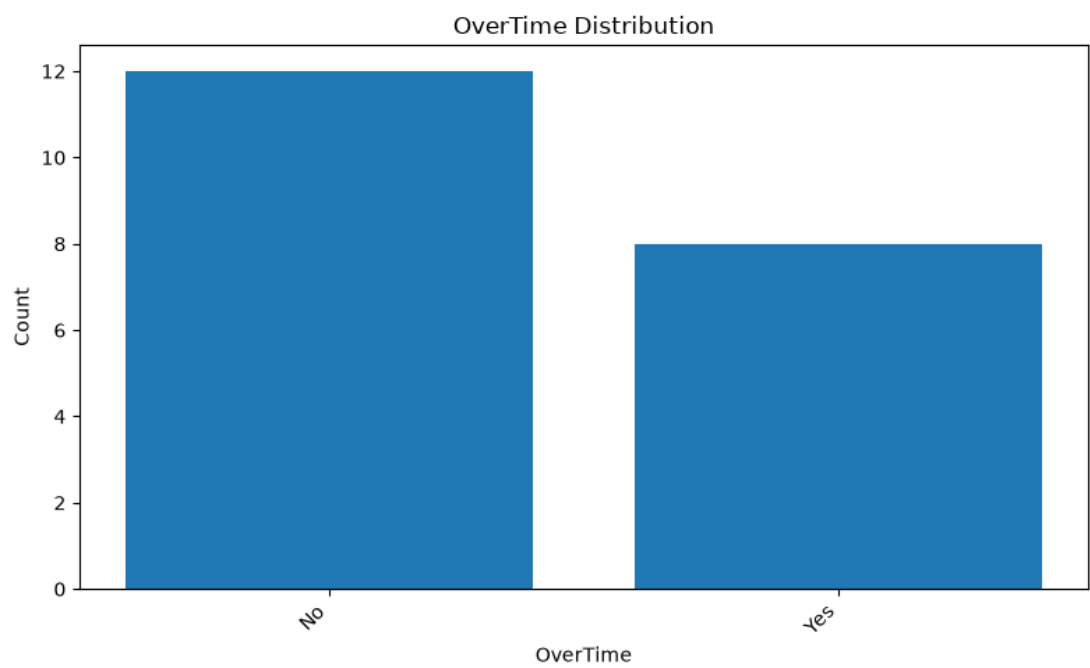
***Education Countplot***



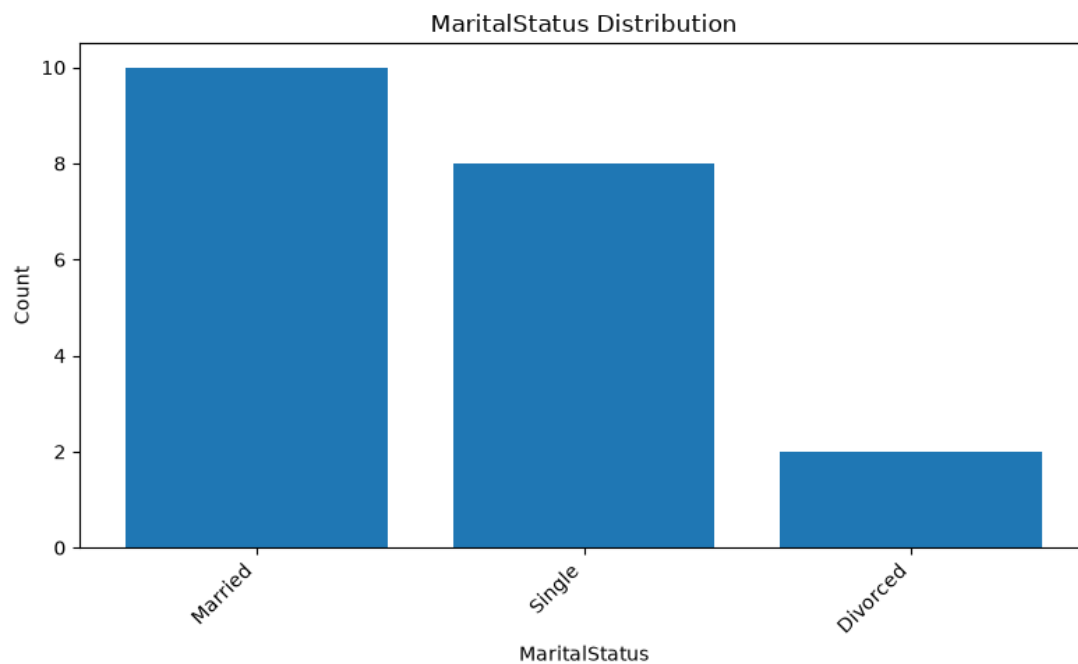
***Worklifebalance Countplot***



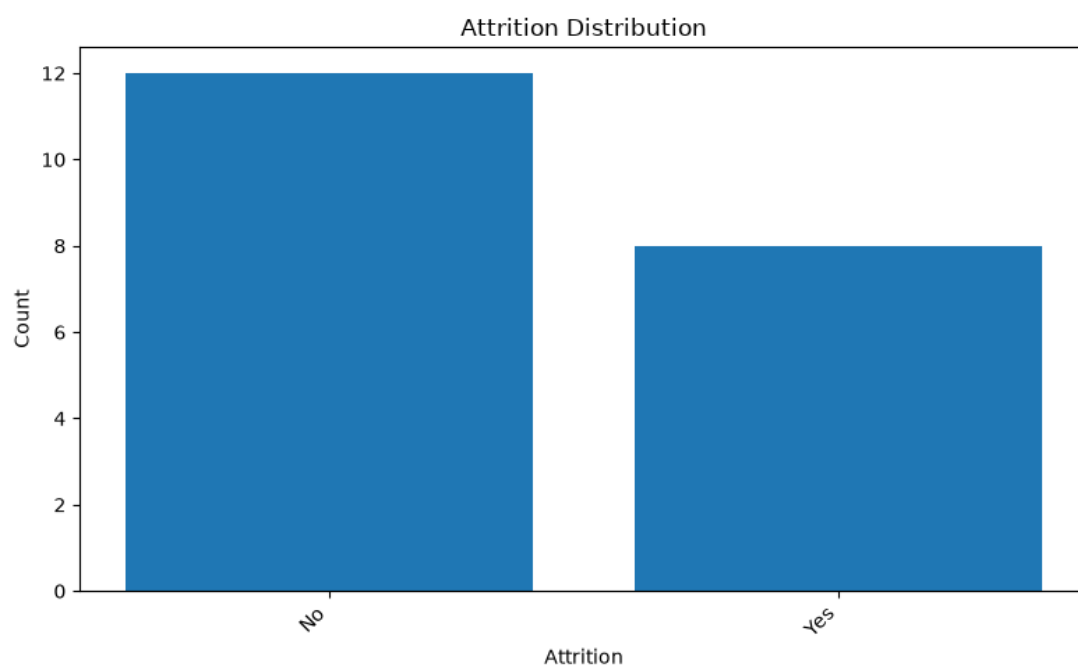
***Overtime Countplot***



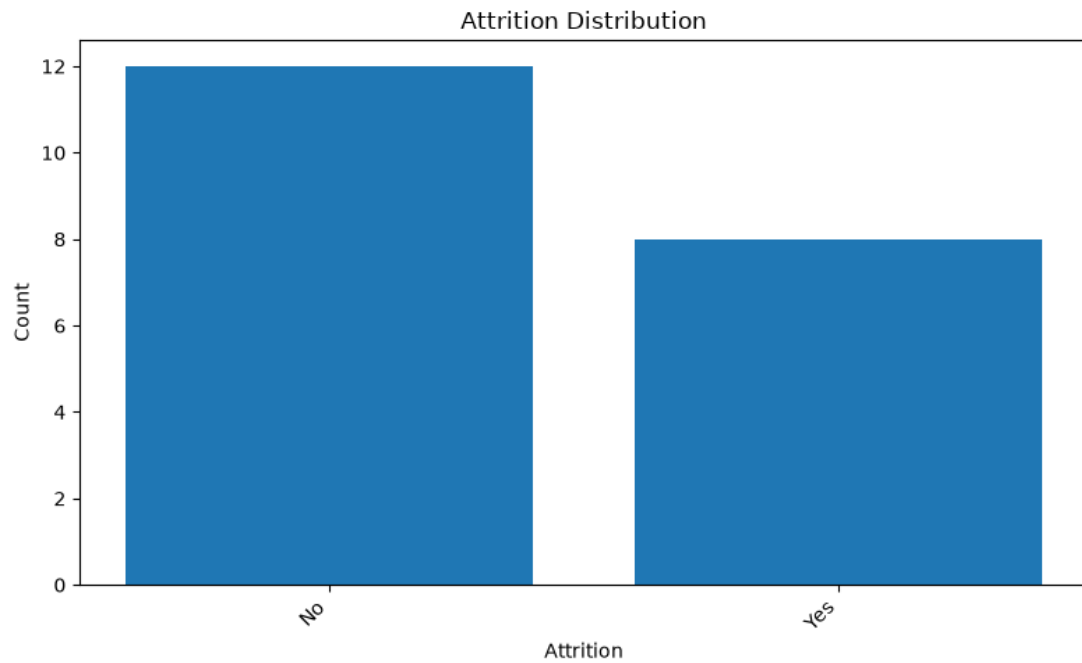
***Maritalstatus Countplot***



***Attrition Countplot***



***Target Distribution***



## 8. Feature Engineering

**Total Features:** 20

**Feature Matrix Shape:** (20, 20)

**Generated Features:**

- Age
- Salary
- YearsAtCompany
- Gender\_Female
- Gender\_Male
- Department\_Finance
- Department\_HR
- Department\_IT
- Department\_Sales
- Education\_Bachelor
- Education\_Master
- Education\_PhD
- WorkLifeBalance\_Excellent
- WorkLifeBalance\_Fair
- WorkLifeBalance\_Good
- OverTime\_No
- OverTime\_Yes
- MaritalStatus\_Divorced
- MaritalStatus\_Married
- MaritalStatus\_Single

## 9. Model Training

| Model               | Score  |
|---------------------|--------|
| logistic_regression | 1.0000 |
| random_forest       | 1.0000 |

**Best Model:** logistic\_regression

**Best Score:** 1.0000

**Training Samples:** 16

**Testing Samples:** 4

## 10. Model Evaluation

| Metric    | Value  |
|-----------|--------|
| accuracy  | 1.0000 |
| precision | 1.0000 |
| recall    | 1.0000 |
| f1_score  | 1.0000 |

### *Confusion Matrix*

| Actual / Predicted | 0 | 1 |
|--------------------|---|---|
| 0                  | 2 | 0 |
| 1                  | 0 | 2 |

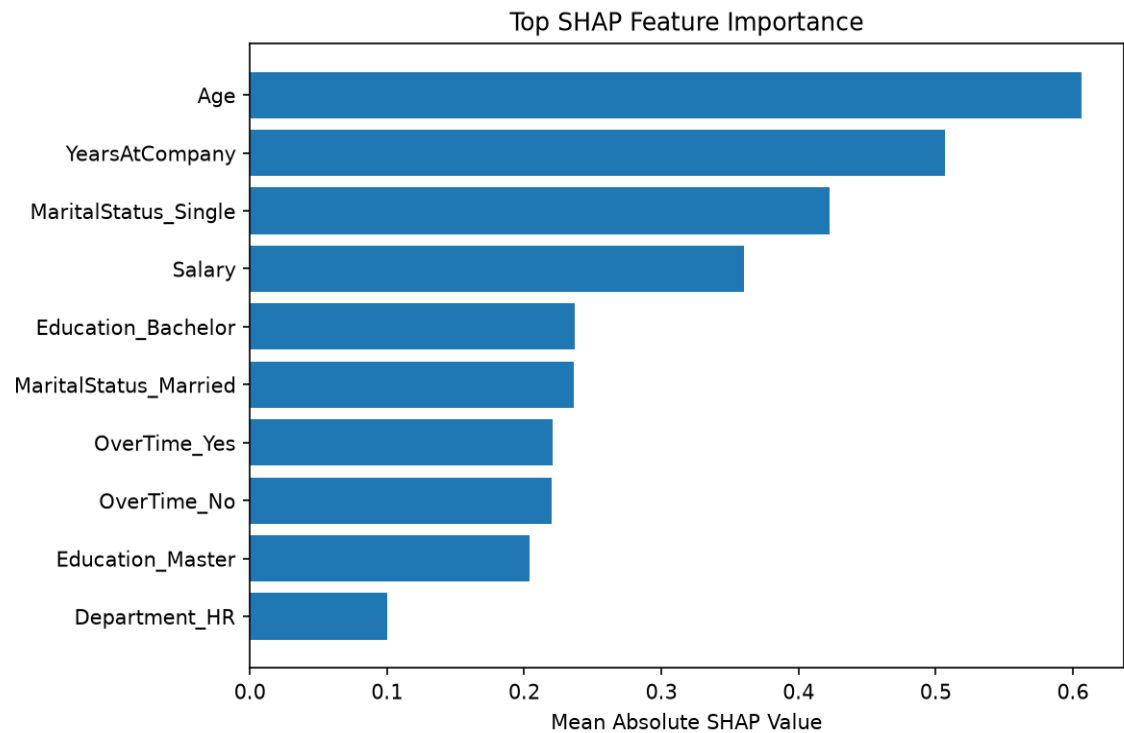
## 11. Model Explainability

### *Top Features*

| Feature               | Importance |
|-----------------------|------------|
| Age                   | 0.6065     |
| YearsAtCompany        | 0.5068     |
| MaritalStatus_Single  | 0.4223     |
| Salary                | 0.3600     |
| Education_Bachelor    | 0.2368     |
| MaritalStatus_Married | 0.2364     |
| OverTime_Yes          | 0.2204     |
| OverTime_No           | 0.2201     |
| Education_Master      | 0.2037     |

|               |        |
|---------------|--------|
| Department_HR | 0.0999 |
|---------------|--------|

SHAP Feature Importance



12. Business Insights

Executive Summary

The machine learning model has achieved near-perfect performance on this small dataset, with an accuracy of 1.0 and perfect precision, recall, F1-score in the classification task to predict employee attrition. However, given the limited size of the dataset (only 20 rows), these results may not generalize well to a larger or more diverse population. The model relies heavily on features such as age, years at company, marital status, and salary, with less emphasis on other factors like department and work-life balance.

Key Findings

- The dataset is small (20 rows) which may limit the model's ability to generalize well.
- Features such as 'age', 'years\_at\_company', and 'marital\_status' are highly influential in determining attrition, while other factors like 'department' have less impact.
- The model performs perfectly on this specific dataset but requires validation with additional data.

Model Performance Summary

The model exhibits perfect performance metrics: accuracy = 1.0, precision = 1.0, recall = 1.0, and F1-score = 1.0. This indicates that the predictions are both highly accurate and balanced across classes, meaning the model is not biased towards predicting one class over another.

Feature Importance Summary

The feature importance analysis shows that 'age', 'years\_at\_company', and 'marital\_status' are critical in determining employee attrition. These features have a strong positive influence on the prediction outcomes. On the other hand, while some factors like 'salary' also play an important role, others such as 'department' do not significantly impact predictions.

### ***Business Recommendations***

- Given the small dataset size, it is recommended to gather additional data for model validation and testing to ensure robustness.
- Focus on retaining employees by addressing key factors identified in this analysis: age, years of service, marital status, and salary.
- Consider implementing targeted retention programs based on these insights.

### ***Risks and Limitations***

- The model's performance may not generalize well to a larger or more diverse population due to the small sample size.
- Overfitting is possible with such limited data; further validation is necessary to avoid this risk.
- The analysis assumes that all features are relevant and correctly interpreted. Additional domain knowledge can help validate these assumptions.

### ***Next Steps***

- Collect more data to improve the model's generalizability and robustness.
- Perform cross-validation techniques to ensure the model's performance is not overly optimistic.
- Analyze real-world retention programs based on identified factors for improved outcomes.